MIL-P-63065A (AR)
18 January 1982
SUPERSEDING
MIL-P-63065 (PA)
10 August 1976

MILITARY SPECIFICATION

PROJECTILE, 155MM: HE, M692/M731
METAL PARTS ASSEMBLY

This Specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements, quality assurance provisions and packaging for Projectile, 155MM, HE, M692/M731 Metal Parts.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-6867	- Magnetic Inspection Units
MIL-I-6868	- Inspection Process, Magnetic Particle
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-410	- Certification of Magnetic Particle Inspection Personnel
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures for Sampling by Attributes

2.1.2 Other government documents, drawings, and publications - The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9298350	- Projectile, 155MM, M692/M731, Metal Parts
8837839	- Pallet for Projectile, 155MM, HE and GB Hollow Base Type
8648916*	- Eddy Current Inspection Req for Base, Dwg. D10542904 part of Projectile 155MM, M483A1
9280318*	- Body Magnetic Particle Standard for 155MM, ICM, M483A1

*Applicable to Projectile 155MM, HE, M692/M731

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence - In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

PUBLICATIONS

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless othewise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

ASTM Method E-8	- Tension Testing of Metallic Materials
ASTM Method E-10	- Brinell Hardness of Metallic Materials

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ASTM Method E-18	- Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM Method E103	- Rapid Indentation Hardness Testing of Metallic Materials
ASTM Method B117	- Salt Spray (Fog) Testing
ASTM Method B247	- Aluminum-Alloy Die and Hand Forgings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

3. REQUIREMENTS

3.1 General - Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications.

3.2 Projectile body

3.2.1 Forming - When a cold working operation is used in forming the projectile body, the maximum reduction in cross sectional area shall not exceed 22 percent. Process drawings showing the prior to cold work and after cold work process piece dimensions shall be submitted to cognizant engineering agency, for approval prior to use. Once approved, the process drawings submitted, shall not be changed without prior approval of the procuring activity. The hardness of the process piece prior to a cold working operation shall not exceed 241 Brinell. (see 6.6)

3.2.2 Stress relief - Prior to the band welding operation, projectile bodies which have been cold worked shall be stress relieved at 1000°F minimum temperature (body) for a minimum time of 1/2 hour.

3.2.3 Heat treatment - Each projectile body (After application of the welded overlay band) shall be austenitized, quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing. No work involving plastic flow is permitted after tempering unless followed by a stress relief. (See note 6.12).

3.3 Rotating band

3.3.1 Rotating band application - The rotating band shall be overlay welded using the gas metal arc welding process in accordance with the applicable drawings and specifications. Gas metal arc welding process in this instance means the use of a consumable electrode with a non-electrode auxiliary wire. The width of the band shall be obtained by oscillating the welding head and the auxiliary wire feed system. The process for applying the rotating band, once established must be maintained, controlled and monitored by the contractor. Procedures for maintaining, controlling and

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monitoring the process must be submitted to the technical agency for approval prior to the start of production. Documented records of the process control must be made available to the Government upon request. (See 6.9).

3.3.2 Rotating band seat - Body material (steel) under the rotating band shall not be visible in the cannelure diameter.

3.3.3 Foreign matter on band seat - Prior to banding, the band seat should be free from any contaminants that would be detrimental to the adhesion of the band to the band seat.

3.4 Torque

3.4.1 Fiberglass wrapping to body - The fiberglass wrapping shall withstand a minimum torque of 550 foot pounds.

3.5 Metal defects

3.5.1 Ogive, body and base - The ogive, body and base shall be free from all metal defects, in accordance with the defect standards (see 3.5.3) including cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects. In addition, the cavity of the components shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, and pits. (See 3.5.3)

3.5.2 Rotating band - The rotating band and band seat areas shall have no cracks, lack of bonding, or other metal defects in accordance with defect standards. (See 3.5.3).

3.5.3 Defect standards - Defect standards must be approved by the technical engineering agency. All visual standards for defects will be established for each contract after the first months production or whenever requested by the procuring contracting agency.

3.6 Proving ground function test - The assembled projectile metal parts shall reveal no evidence of failure when tested at excess and service pressure. Metal parts failure shall be:

A. Loss or breakup of metal part components in flight (except obturator).

B. Shear of the rotating band as evidenced by complete lack of engraving.

C. Loss or loose fiberglass wrapping.

D. Base plug and cargo not ejecting (except Zone 8SX) (Failure analysis must determine the cause to be the projectile metal parts).

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3.7 Ogive and base -

3.7.1 Microstructure - The microstructure of the Ogive and Base shall comply with the MICROSTRUCTURE depicted in Figure 1a or 1b.

3.7.2 Heat treatment - Each base and ogive shall be solution heat treated and aged uniformly to meet the mechanical properties specified on the applicable drawing.

3.7.3 Submission of base forging drawing - Prior to the initiation of production, a drawing of the Base Forging will be submitted to the Procurement Contracting Officer for approval.

3.8 First article inspection - This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.9 Control dimensions -

3.9.1 Tool control dimensions - Dimensions marked tool control shall be gaged at intervals determined by the contractor. The contractor will submit to the technical agency a plan for inspecting tool control dimensions. Production shall not begin prior to approval of this plan. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor. The procedure shall be submitted to the technical agency for approval prior to use.

3.9.2 Process control dimensions - The contractor shall prepare a process control plan for the dimensions marked process control. The plan shall be submitted to the technical agency for approval prior to use.

3.10 Records of results of contractor conducted examinations and tests - The contractor shall maintain records of the results of all examinations and tests required by this specification and submit same upon request by the Government. (See 6.4).

3.11 Workmanship - The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Burr and sharp edges - No component shall have a burr or sharp edge which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

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3.11.2 Foreign matter - All parts and assemblies shall be free of chips, dirt, grease, oil, rust, or other foreign matter which may interfere with final inspection and completely free of same prior to shipping.

3.11.3 Threads - Threads shall be full and undamaged, for the entire minimum length or depth as required by the applicable drawings.

3.11.4 Painting - Painting shall comply with the requirements of the applicable drawing. Drying time prior to testing shall be in accordance with the applicable drawing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections - The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission - The contractor shall submit a first article as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities as indicated.

PART DESCRIPTION	DRAWINGS	QUANTITY
Projectile metal parts assembly complete with bases	9298350	25
Projectile subassembly	9347119	5
Body, wrapped	9298341	5
Body	9298354	5
Body (prior to banding)	9298354	2
Ogive	9298324	5
Ogive forging (Prior to heat treat)	-	1
Base (prior to hardcoat and drilling wrench hole)	9298335	5

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Base Hardcoated	9298335	5
Base Forging (Prior to heat treat)	-	1
Obturator	10542907	5

Twenty of the above projectile metal parts assemblies and bases shall be forwarded to a designated load plant for loading and shall then be forwarded to a government proving ground for ballistic testing.

4.3.2 Inspections to be performed - See MIL-A-48078 and Table 1 herein.

4.3.3 Rejection - See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Projectile Metal Parts for Projectile, 155MM, HE, M692/M731				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ogive				
	(Dwg 9298324)				
	Examination for Defects	5	-	3.1	4.4.2.1
	Salt Spray Test	1	-	3.1	4.5.8.1
	Yield Strength & Percent Elongation	2 Specimens	-	3.7.2	4.5.1.5
	Microstructure	1	-	3.7.1	4.5.6
	Obturator				
	(Dwg 10542907)				
	Examination for Defects	5	-	3.1	4.4.2.2
	Base				
	(Dwg 9298335)				
	Examination for Defects	5	-	3.1	4.4.2.3
	Yield Strength & Percent Elongation	2	-	3.1	4.5.1.5
	Microstructure	1	-	3.7.1	4.5.6

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION (Cont'd)

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Projectile Metal Parts for Projectile, 155MM, HE: M692/M731				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body (Dwg 9298354)				
	Examination for Defects	5	-	3.1	4.4.2.6
	Yield Strength & Percent Elongation	2	-	3.2.3	
	Body Wrapped (Dwg 9298341)				
	Examination for Defects	5	-	3.1	4.4.2.6
	Torque Test of Fiberglass Wrap	2	-	3.4	4.5.9
	Fiber Glass Content	2	-	3.1	4.5.10
	Projectile Sub-Assembly (Dwg 9347119)				
	Examination for Defects	5	-	3.1	4.4.2.7
	Projectile Metal Parts (Painted) (Dwg 9298350)				
	Examination for Defects	25	-	3.1	4.4.2.8
	Salt Spray Test	2	-	3.1	4.5.8.1
	Ballistic Tests		-	3.6	4.5.12
	A. Excess Pressure	10	-	3.6	4.5.12.1
	B. Zone 3 ($-40^{o} \pm 5^{o}F$)	5	-	3.6	4.5.12.2
	C. Service Pressure (7 Foot Drop Test)	5	-	3.6	4.5.12.3

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectiles shall contain:

a. Ogives from not more than one forging manufacturer.
b. Bases from not more than one forging manufacturer.
c. Bodies, from not more than one forging manfuacturer.

4.4.1.1 Body heat treatment lot - A heat treatment lot of bodies is defined as follows:

4.4.1.1.1 Standard procedure - A heat treatment lot of bodies shall consist of bodies from only one mill heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature and for the same time cycles, in one unchanged process.

4.4.1.1.1.1 Miscellaneous lot - Bodies manufactured from small heats of steel, or bodies that have become separated from previously delivered lots may be grouped into a miscellaneous heat treatment lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1.1. The maximum lot size shall be 2000 bodies.

4.4.1.1.2 Use - The heat treatment lot constituted per 4.4.1.1.1, 4.4.1.1.1.1 shall be the basis for selecting samples for mechanical properties tests. Bodies in each heat treatment lot shall be identified to that particular lot until satisfactory completion of mechanical properties tests.

4.4.1.2 Heat treatment lot (ogive or base) - A heat treatment lot of ogives or bases shall consist of components manufactured from material of the same alloy and temper from one manufacturer. The material shall be heat treated by the same process, in the same equipment and using the same temperature and time cycles. At no time will the lot quantity exceed 4800.

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The heat treatment lot thus constituted shall be the basis for selecting samples for mechanical properties tests. Components in each heat treatment lot shall remain segregated and identified until satisfactory completion of mechanical properties tests.

4.4.1.3 - Miscellaneous lot (ogive or base) - Components manufactured from small lots of material or those that have become separated from previously heat treated lots may be grouped into a miscellaneous lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.2. The lot thus constituted shall be the basis for selecting samples for the mechanical properties tests.

4.4.1.4 Projectile lot - Heat treatment lots that have been determined to have complied with the yield strength and percent elongation requirements of the applicable drawing may be regrouped for assembly into projectile lots.

4.4.2 Examination - See MIL-A-48078.

a. Sampling plans - Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-1235 or MIL-STD-105 as applicable.

b. Automatic gaging - Facilities equipped with government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with contract.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER 9298324
4.4.2.1	Ogive		NEXT HIGHER ASSEMBLY 9347119

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
SPECIAL					
A	Metal Defective		100%	3.5	Visual
B	Hydrostatic	See 4.5.5		3.1	4.5.5
MAJOR					
101	Pitch diameter of external thread, min.		.40%	3.1	Gage
102	Major diameter of external thread, min.		.40%	3.1	Gage
103	Diameter of cavity, rear		.40%	3.1	Gage
104	Length from rear face to forward inside basic diameter		.40%	3.1	Gage
105	Length from rear face rear inside basic diameter		.40%	3.1	Gage
106	Variation in wall thickness		.40%	3.1	Gage
107	Pitch diameter of fuze thread		.40%	3.1	Gage
108	Minor diameters of fuze thread, max.		.40%	3.1	Gage
109	Second diameter rear of fuze thread undercut		.40%	3.1	Gage
110	First diameter rear of fuze thread undercut		.40%	3.1	Gage
111	Depth from front face to start of second diameter rear of fuze thread undercoat		.40%	3.1	Gage
112	Length from front face to forward basic O.D.		.40%	3.1	Gage
113	Length from front face to rear basic O.D.		.40%	3.1	Gage
114	Length of fuze thread, min		.40%	3.1	Gage
115	True position of second diameter rear of fuze thread undercut with pitch diameter of fuze thread		.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	Ogive				9298324
					NEXT HIGHER ASSEMBLY 9347119
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116	True position at rear of fuze thread undercut with pitch diameter of fuze thread		.40%	3.1	Gage
117	Depth from rear face to bottom of cavity		.40%	3.1	Gage
118	Runout of outer ogive surface to base thread pitch diameter and shoulder (2 places)		.40%	3.1	Gage
119	Runout of fuze thread pitch diameter projections (2) to body shoulder and fuze mating surface		.40%	3.1	Gage
120	Yield Strength and Percent Elongation	**	-	3.7.2	4.5.1.5
121	Microstructure	**	-	3.7.1	4.5.6
122	Protective Finish	**	-	3.1	4.5.8.1
123	Diameter of lock ring groove		.40%	3.1	Gage
124	Perpendicularity rear face with pitch diameter of threads		.40%	3.1	Gage
MINOR					
201	Diameter of body thread undercut		.65%	3.1	Gage
202	Length from front face to shoulder		.65%	3.1	Gage
203	Depth from front face to rear of fuze thread undercut		.65%	3.1	Gage
204	Diameter of fuze thread undercut		.65%	3.1	Gage
205	Width of fuze thread undercut		.65%	3.1	Gage
206	Length, total		.65%	3.1	Gage
207	Radii of chamfers missing or incorrect		.65%	3.1	Visual
208	Surface finish improper		.65%	3.1	Visual
209	Burr		.65%	3.11	Visual
210	Foreign matter		.65%	3.11	Visual
211	Evidence of poor workmanship		.65%	3.1	Visual

NOTES:

**Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Obturator		10542907
			NEXT HIGHER ASSEMBLY 9298350

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	True position of rear I.D. with forward I.D.		.40%	3.1	Gage
102	Diameter outside, front		.40%	3.1	Gage
103	Diameter outside, rear		.40%	3.1	Gage
104	Diameter inside, front		.40%	3.1	Gage
105	Depth of forward inside diameter		.40%	3.1	Gage
106	Thickness of wall at rear		.40%	3.1	Gage
107	Length, total		.40%	3.1	Gage
MINOR					
201	Radii or chamfers missing or incorrect		.65%	3.1	Visual
202	Evidence of poor workmanship		.65%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.3	Base Without Cap, Protector		9298335
			NEXT HIGHER ASSEMBLY 9298350

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
a	Metal Defective		100%	3.5	4.5.11
Major					
101	Runout forward of taper with pitch diameter of threads and shoulder		.40%	3.1	Gage
102	Runout rear of taper with pitch diameter of threads and shoulder		.40%	3.1	Gage
103	Perpendicularity of shoulder with pitch diameter of thread		.40%	3.1	Gage
104	Diameter of cavity		.40%	3.1	Gage
105	Length from shoulder to forward basic diameter		.40%	3.1	Gage
106	Length from shoulder to rear basic diameter		.40%	3.1	Gage
107	Length from front face to shoulder		.40%	3.1	Gage
108	Length of thread		.40%	3.1	Gage
109	Thickness through base web		.40%	3.1	Gage
110	Pitch diameter of thread		.40%	3.1	Gage
111	Major diameter of thread, min.		.40%	3.1	Gage
112	Thread(s) damaged or not full		.40%	3.1	Gage
113	Yield Strength and Percent Elongation	*		3.7	4.5.1.5
114	Protective Finish	*		3.11	4.5.8.1 and 4.5.8.2
115	Flatness of front face		.40%	3.1	Gage
116	Parallelism of front face with shoulder		.40%	3.1	Gage

NOTES:

*Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.3	Base Without Cap, Protector		9298335
			NEXT HIGHER ASSEMBLY 9298350

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	Diameter of thread pilot		.65%	3.1	Gage
202	Diameter of thread undercut		.65%	3.1	Gage
203	Length, total		.65%	3.1	Gage
204	Length from front face to bottom of wrench holes, min.		.65%	3.1	Gage
205	Width of thread undercut		.65%	3.1	Gage
206	Presence and location of wrench holes		.65%	3.1	Visual/Func. Gage
207	Radii or chamfers missing or incorrect		.65%	3.1	Visual
208	Surface finish improper		.65%	3.1	Visual
209	Protective coating inadequate or damaged		.65%	3.5	Visual
210	Burr		.65%	3.11	Visual
211	Foreign matter		.65%	3.11	Visual
212	Protective coating on threads, or indicated surface		.65%	3.11	Visual
213	Evidence of poor workmanship		.65%	3.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298335
4.4.2.4	Base with Cap, Protector *				NEXT HIGHER ASSEMBLY 9298350
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Protector cap missing		.65%	3.1	Visual
202	Evidence of poor workmanship		.65%	3.1	Visual

NOTES:

*Not required for intraplant shipment.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298354
4.4.2.5	Body (prior to banding)				NEXT HIGHER ASSEMBLY 9298341
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Metal defective in band seat area		100%	3.5.2	Visual
MAJOR					
101	Diameter of rotating band seat		.40%	3.1	Gage
102	Location of rotating band seat		.40%	3.9.2	4.5.2
MINOR					
201	Radii or chamfers missing or incorrect		.65%	3.1	Visual
202	Burr		.65%	3.11	Visual
203	Foreign matter on band seat		.65%	3.3.3	Visual
204	Surface finish of band seat improper		.65%	3.1	Visual
205	Evidence of poor workmanship		.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Body and Body Wrapped	1 OF 4	9298354/9298341
			NEXT HIGHER ASSEMBLY 9347119

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Metal Defective				
	a. Body		100%	3.5	4.5.3.2
SPECIAL					
A	Band seat metal defective	*		3.5.2	4.5.3.1
MAJOR					
101	Hardness Test of Body		100%	3.2	4.5.1.2
102	Yield Strength and percent elongation	*		3.2	4.5.1.4
103	Band Chemical Analysis	*		3.1	4.5.4
104	Torque of Fiberglass Wrapping to Body	*		3.4	4.5.9
105	Fiberglass Content	*		3.1	4.5.10
106	Runout of body undercut with bourrelets before fiberglass wrap		.40%	3.1	Gage
107	Runout of forward body ogive with bourrelets		.40%	3.1	Gage
108	Runout of rear body ogive with bourrelets		.40%	3.1	Gage
109	Runout of rotating band lip with bourrelets		.40%	3.1	Gage
110	Runout of fiberglass wrapping with bourrelets		.40%	3.1	Gage
111	Runout of pitch diameter of base thread with bourrelets and rear face		.40%	3.1	Gage
112	Runout of rear outside diameter with bourrelets		.40%	3.1	Gage
113	Runout of body cavity with bourrelets		.40%	3.1	Gage
114	True position of body ogive thread counterbore with minor diameter of body ogive thread		.40%	3.1	Gage

NOTES:

*Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 4		DRAWING NUMBER
4.4.2.6	Body and Body Wrapped				9298354/9298341
					NEXT HIGHER ASSEMBLY
					9347119
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MAJOR					
115	Pitch diameter of body ogive thread, max.		.40%	3.1	Gage
116	Minor diameter of body ogive thread, max.		.40%	3.1	Gage
117	O.D. of undercut (before fiberglass)		.40%	3.1	Gage
118	Length to forward body ogive basic diameter		.40%	3.1	Gage
119	Length to rear body ogive basic diameter		.40%	3.1	Gage
120	Length total		.40%	3.1	Gage
121	Length from rear face to front of rear bourrelet		.40%	3.1	Gage
122	Length from rear face to rear of forward bourrelets		.40%	3.1	Gage
123	Location of forward cannelure groove		.40%	3.1	Gage
124	Location of rear cannulure groove		.40%	3.1	Gage
125	Perpendicularity of rear face with pitch diameter of base thread		.40%	3.1	Gage
126	O.D. of bourrelets, min. (front & rear)		.40%	3.1	Gage
127	O.D. of fiberglass wrappings, min.		.40%	3.1	Gage
128	O.D. of rotating band lip		.40%	3.1	Gage
129	O.D. of rotating band flats		.40%	3.1	Gage
130	O.D. of rotating band cannelures		.40%	3.1	Gage
131	O.D. of obturator seat		.40%	3.1	Gage
132	I.D. of body cavity beyond tapered section		.40%	3.1	Gage
133	O.D. rear of body		.40%	3.1	Gage
134	Depth of body cavity keyway	*	.40%	3.1	Gage
135	Location to bottom of body cavity keyway	*	.40%	3.1	Gage

NOTES: *Alternates

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Body and Body Wrapped	3 OF 4	9298354/9298341
			NEXT HIGHER ASSEMBLY: 9347119

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MAJOR (cont)					
136	Length of body cavity		.40%	3.1	Gage
137	Length of body cavity keyway from rear face		.40%	3.1	Gage
138	Length from rear face to rear of rotating band		.40%	3.1	Gage
139	Thickness of wall from bottom of keyway to outer surface, at forward end of body cavity		.40%	3.1	Gage
140	Thickness of wall at forward end of cavity		.40%	3.1	Gage
141	Width of rotating band		.40%	3.1	Gage
142	Width of rotating band relief groove		.40%	3.1	Gage
143	Width of body cavity keyway		.40%	3.1	Gage
144	Band seat visible in rotating band cannelures		.40%	3.3.2	Visual
145	Pitch diameter of base thread		.40%	3.1	Gage
146	Minor diameter of base thread, max.		.40%	3.1	Gage
147	True position of base thread counterbore with pitch diameter of base thread		.40%	3.1	Gage
148	Length of effective base thread, min.		.40%	3.1	Gage
149	Length from front of band to band basic diameter		.40%	3.1	Gage
150	Runout of pitch diameter of ogive & thread to bourrelets and ogive mating surface		.40%	3.1	Gage
151	Runout of tapered section		.40%	3.1	Gage
152	Location of rear I.D. basic diameter		.40%	3.1	Gage
153	Distance between I.D. basic diameters		.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Body and Body Wrapped	4 OF 4	9298354/9298341
			NEXT HIGHER ASSEMBLY: 9347119

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	Diameter of body ogive thread counterbore		.65%	3.1	Gage
202	Depth of body ogive thread counterbore		.65%	3.1	Gage
203	Diameter of base thread counterbore		.65%	3.1	Gage
204	Depth of base thread counterbore		.65%	3.1	Gage
205	Radii or chamfers missing or incorrect		.65%	3.1	Visual
206	Presence and size of undercut forward of rotating band (if applicable)		.65%	3.1	Visual-Gage
207	Surface finish improper		.65%	3.11	Visual
208	Burr		.65%	3.11	Visual
209	Foreign matter		.65%	3.11	Visual
210	Length to rear of fiberglass undercut, min.		.65%	3.1	Gage
211	Length to front of fiberglass undercut, max.		.65%	3.1	Gage
212	Protective Coating inadequate or damaged		.65%	3.1	Gage
213	Evidence of poor workmanship		.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Projectile, Subassembly (before paint)	1 OF 1	9347119
			NEXT HIGHER ASSEMBLY: 9298350

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Metal defective stamp missing (indicating failure to pass test)		100%	3.1	Visual
2	Metal defective		100%	3.5	Visual (probe light)
MAJOR					
101	Weight		100%	3.1	Approved Scale
102	Runout of pitch diameter of fuze thread with bourrelets and front face		.40%	3.1	Gage
103	Runout forward of ogive with bourrelets		.40%	3.1	Gage
104	Runout rear of ogive with bourrelets		.40%	3.1	Gage
105	Gap between ogive and body		.40%	3.1	Gage
106	Length total, max.		.40%	3.1	Gage
107	Air leak test		100%	3.1	4.5.7
MINOR					
201	Surface finish, improper		.65%	3.11	Visual
202	Burr		.65%	3.11	Visual
203	Foreign matter		.65%	3.11	Visual
204	Protective coating inadequate or damaged		.65%	3.11	Visual
205	Evidence of poor workmanship		.65%	3.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.8	Projectile, Metal Parts (after paint)		9298350
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
SPECIAL					
a.	Ballistic Test				
	A. Excess pressure	5		3.6	4.5.12
	B. Zone 3	3		3.6	4.5.12
	C. Service pressure (7 foot drop test)	2		3.6	4.5.12
MAJOR					
101	Protective finish		**	3.11.4	4.5.8.1
102	Diameter of bourrelets, max.		.40%	3.1	Gage
103	Rotating band damaged		.40%	3.1	Visual
104	Fiberglass wrapping damaged		.40%	3.1	Visual
105	Obturator missing or damaged		.40%	3.1	Visual
106	Threads damaged		.40%	3.11.3	Visual
107	Protective finish incomplete or damaged		.40%	3.11.4	Visual
MINOR					
201	Paint on rotating band bearing surfaces in excess of drawing allowances		.65%	3.1	Visual
202	Paint on threads		.65%	3.1	Visual
203	Paint on obturator		.65%	3.1	Visual
204	Paint not dry		.65%	3.1	Visual/Manual

NOTES:

**Sample size as specified in applicable test method paragraph.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9298350
4.4.2.8	Projectile, Metal Parts (after paint)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
205	Marking missing, incorrect or illegible		.65%	3.1	Visual
206	Burr		.65%	3.11	Visual
207	Foreign matter		.65%	3.11	Visual
208	Grommet missing or loose (See note 2)		.65%	3.1	Visual/Manual
209	VCI paper missing (See note 2)		.65%	3.1	Visual/Manual
210	Protective Caps (Base & Fuze Thds) missing		.65%	3.1	Visual
211	Evidence of poor workmanship		.65%	3.11	Visual

NOTES:

Note 2: Not required for intraplant use.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63065A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Pallet*		8837839
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Pallet damaged (unserviceable)		.65%	3.1	Visual
202	Improperly packed		.65%	3.1	Visual
203	Strapping missing, loose or broken		.65%	3.1	Visual/Manual
204	Marking missing, incorrect or illegible		.65%	3.1	Visual
205	Evidence of poor workmanship		.65%	3.1	Visual

NOTES:

*Not required for intraplant use.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing - Testing is described in the First Article Table, Quality Conformance Inspection Tables, and herein.

4.4.3.1 Hardness prior to cold forming - When the contractor elects to cold form the body he will assure compliance with paragraph 3.2.1. Bodies subjected to this test will be hardness tested at a midpoint along the length of outside diameter of the body. For a minimum of 20,000 bodies, each body will be tested. Based upon conformance with the requirement during this testing the contractor may propose an alternate hardness sampling procedure. As a minimum the alternate hardness sampling procedure shall include hardness testing of the first twenty consecutive bodies out of slow cool after each start up and sampling for hardness on a specific hourly basis. The alternate procedure shall be submitted to the procuring activity together with supporting hardness data for approval.

4.4.3.2 Verification of base heat treat - After heat treat and prior to hardcoat each base shall be verified for its having been heat treated. Bases failing to comply will be rejected. Rejected parts may be heat treated and resubmitted. All bases passing this test shall be stamped in accordance with applicable drawing. Testing shall be in accordance with para. 4.5.1.3.

4.4.4 Inspection equipment - For inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.4.1 Automatic inspection equipment - All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification and/or calibration check during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of item produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

4.5 Methods of inspection

4.5.1 Mechanical properties

4.5.1.1 Hardness prior to cold forming - Test shall be performed using the procedure of ASTM Method E-103. Bodies failing to comply with 3.2.1 may be reannealed and retested.

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4.5.1.2 Hardness (body) - In order to determine uniformity of heat treatment each body after application of the welded overlay band and heat treatment and before final machining shall be subjected to a hardness test with a Brinnel Hardness Tester or any other method approved by the procuring activity. Readings shall be taken on the body approximately 3/4 inch forward of the rotating band. The test shall be made using the procedure for performing this test as prescribed by ASTM Method E-10.

4.5.1.3 Verification of heat treat of base - Testing shall be with eddy current or any other approved method.

4.5.1.4 Yield strength and percent elongation of body - The hardest and the softest body as determined by the hardness test in 4.5.1.2 from each heat treatment lot shall be subjected to this test. Two tension test specimens conforming to the largest obtainable round type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of each body, midwall of the band seat area, and two specimens as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of each body, midwall of the area of the body one inch from the front end. The number of specimens and their location are for the convenience of the contractor and government and shall not restrict the contractor or government from testing specimens at other locations to determine that the mechanical properties of the body meet the drawing requirements. The criteria for performing the tests shall be as prescribed by ASTM Method E-8 except elongation shall be measured to the nearest thousandth inch prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth percent.

4.5.1.4.1 Standard lotting procedure - Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the contractor, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be reheat treated to fall within that range or mechanical properties may be taken per 4.5.1.4 from newly selected projectile bodies which represent the true extremes of the hardness range in the heat treatment lot.

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4.5.1.5 Yield strength and percent elongation of ogive and base.- The sample size for the tension test shall be one component (each base and ogive) from each 1200 parts or portion thereof from each heat treatment lot. Two longitudinal test specimens conforming to the largest obtainable round specimen as prescribed by ASTM Method E-8 shall be taken from the sidewall of each sample. The base specimens shall be taken from midwall centered between rear and front ends 180° apart. The ogive tension test specimen shall be taken from the midwall at a location where the gage length center of the specimen is approximately 1 inch from the finished rear face of the body thread end of the ogive, 180° apart. The criteria for performing the tests shall be as prescribed by ASTM Method E-8 except the elongation shall be measured to the nearest thousandth prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth percent. Failure of any specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the quantity represented by the sample.

4.5.2 Rotating band seat location - The contractor may use any control method that will assure the band seat meets the dimensional requirements of the applicable drawing. The inspection plan must include all the required criteria to assure the dimensional aspects of the drawing will be equal to the characteristics defined in 4.4.2.5. When the contractor establishes a control method, the manufacturing procedure, and inspection plan shall be forwarded to the Technical Agency for approval prior to use. No change to these controls will be made once established without approval from the Technical Agency.

4.5.3 Inspection for body metal defects after machining

4.5.3.1 Band seat - The first five banded bodies, after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to this test. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been visually examined and found acceptable. Reduced sampling of one in each 8 hours production or fraction thereof per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been visually examined and found acceptable. (see 6.9)

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The samples provided shall have the band removed by machining the band and chemically etching without damaging the base metal (steel) of the band seat (see 6.7). Each body sampled shall then be inspected using the magnetic particle inspection technique.

The procedure for performing this test shall be as specified in MIL-I-6868 for wet fluorescent method with an indirect magnetization by either the continuous or residual method, inducing a circular magnetic field, and again after inducing a longitudinal magnetic field.

Should two consecutive failures occur in any welding machine(s), in any reduced sample size, the machine(s) shall revert back to the original first five consecutive banded bodies per that welding machine(s) and after five consecutive acceptable bodies, the sequence of reduced sampling will also apply. Banded bodies shall be identified along with the welding machine that applied the band.

4.5.3.2 Finished body - Projectile bodies after final machining and prior to fiberglass wrap shall be inspected by the magnetic particle inspection technique. The procedures for performing this test shall be as specified in MIL-I-6868, for wet fluorescent method with indirect magnetization by either the continuous or residual methods. The entire body, inside and outside, shall be inspected after inducing a circular magnetic field and again after inducing a longitudinal magnetic field. The magnetic particle standard per Dwg. 9280318 shall be utilized to establish inspection sensitivity. Any body which fails to comply with the requirements of 3.5 shall be rejected. Bodies which comply shall be stamped with the letter "M" as specified on the applicable drawing.

4.5.3.3 Magnetic particle suspension - The solid content of the suspension shall be maintained at the level recommended by the suspensoid manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by Section 4 of MIL-I-6868. When performing this test for solid content of the suspension, if it is determined that it does not comply, all bodies that have been tested and accepted per 4.5.3.2 since the last satisfactory test was performed shall be retested after the suspension has been brought to the proper level. The magnetic particle test shall be performed on equipment that complies with the requirements of specification MIL-M-6867. The operator(s) of the equipment and the inspector(s) making the determinations shall be certified in accordance with MIL-STD-410.

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4.5.4 Band chemical analysis - The first five banded bodies, after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to this test. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been tested and found acceptable. Reduced sampling of one in each 4 hours production or fraction thereof per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been tested and found acceptable. Sample chips shall be obtained from each band from the last .050 inches removed when machining to the finished diameter. All chips shall be collected and identified with the shell from which the chips were machined. A minimum 5 gram sample from each complete specimen thus obtained from each band shall be dissolved and the analysis made from the individual solutions. The sample may not be cleaned with a magnet.

If a failure occurs sampling will revert to the original first five consecutive banded bodies per that machine(s) and after five consecutive acceptable bodies, the sequence of reduced sampling will also apply. Failure to comply with the requirements of the drawing shall be cause for rejection of the production represented by the sample.

4.5.5 Hydrostatic test of ogive -Prior to the application of protective finish, the finish machined ogive shall be subjected to a hydrostatic test. The ogive shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours of continuous operation. It shall also be calibrated prior to re-starting the operation after a layoff of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the test pressure was actually lower than that specified on the applicable drawing, the ogives tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. The fixture should be such that it does not restrain the ogive. That is, it shall simulate actual assembly conditions.

4.5.5.1 Lot sampling plan - 100% hydrostatic testing of finish machined ogives shall be conducted until 1200 ogives consecutively pass the hydrostatic test. Subsequently lots of no greater than 5000 ogives may be submitted for testing. A sample of 50 ogives per lot shall then be tested. Failure of any ogive to meet the requirement of the applicable drawing shall be cause for rejection of the lot represented by the sample. When a lot has been rejected while on sampling inspection, the frequency shall revert back to the 100 percent inspection until 1200 ogives consecutively pass the hydrostatic test, to qualify for reduced sampling testing.

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4.5.5.2 Continuous sampling (option) - The ogive hydrostatic testing may be conducted using a continuous sampling plan from MIL-STD-1235, CSP-1 with AQL of 0.10.

4.5.6 Microstructure examination (ogive & base) - One ogive & one base forging from each heat treatment lot shall be selected for microstructure examination. The metallographic specimen shall be etched with Keller's Reagent and all areas examined under a microscope/metallograph at a minimum magnification of 250 diameters. Any evidence of the insoluble phase generally occurring in a semi-continuous or continuous network form shall be cause for rejection of the lot represented by the sample.

Two examples of acceptable microstructures are shown in Figures 1a & 1b. In Figure 1a, the insoluble phase is well broken-up randomly distributed within a matrix which is essentially unrecrystallized. In Figure 1b, the insoluble phase is well broken-up and randomly distributed within a matrix which is partially recrystallized. Two examples of the unacceptable microstructures are shown in Figure 2a and 2b. In Figure 2a, the insoluble phase is distributed in a semi-continuous form within the matrix which shows microsegregation (or coring). In Figure 2b, the insoluble phase is distributed in a more continuous form (network) within a matrix which shows microsegregation (or coring). Either of the above structures (Figure 2a & 2b) must be avoided because they are detrimental to the integrity of the material.

Microstructural preparation and examination shall be performed by a metallographer or metallurgical engineer familiar with aluminum alloys.

4.5.6.1 Ogive specimen: A metallographic specimen at least 1/2 inch in length shall be removed from the sidewall. It shall be a longitudinal cross section of the body thread area with the center of the specimen approximately 3/4 inches from the finished rear face.

4.5.6.2 Base specimen: A metallographic specimen at least 1/2 inch in length shall be removed from the base and be a longitudinal cross section with its center located at mid-diameter of the base.

4.5.7 Air leak test - The ogive to body joint of the projectile subassembly shall be tested in accordance with the applicable drawing for 15 seconds minimum. A force shall not be applied in the axial direction such as to compress the ogive to body joint. Failure to comply shall be cause to reject the projectile subassembly. Projectile subassemblies which comply shall be stamped with the letter A in 1/8 inch letters in the area indicated on the drawing. The test may be accomplished by one of the following procedures.

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4.5.7.1 Submersion test (alternate 1) - Metal parts subassemblies shall be mounted in an approved fixture in such a manner that the subassembly shall be submerged in water. The required air pressure shall be applied and held for the minimum specified time, after the water in the tank is permitted to settle after immersion. Observation for leakage shall be made for the entire time the metal parts subassembly is required to be pressurized. The presence of air bubbles in the vicinity of the body to ogive joint or rising to the surface of the water shall be cause for rejection of the metal parts subassembly.

4.5.7.2 Pressure drop test (alternate 2) - Each projectile sub-assembly shall be mounted so as to seal the projectile above and below the ogive to body joint with a collar (5 in^3 max.) and pressurizing this collar at 15 psi for 15 seconds while looking for a maximum pressure drop of 0.01 psi during the 15 second hold cycle. Pressure source shall be inactive during the hold cycle. A pressure drop in excess of 0.01 psi shall be cause for rejection of the subassembly.

4.5.8 Protective finish test

4.5.8.1 Salt spray test - The salt spray test shall be performed on each of the following; projectile subassembly, ogive and base utilizing the equipment, methods and procedures specified in ASTM B-117 and in accordance with the applicable detailed specification for the protective finish, except the curing cycle for the body assembly shall be in accordance with dwg. 9298350. Testing shall be performed on one ogive and base for each eight hours of production or fraction thereof. The sample size for the subassembly shall be as specified in the applicable specification. Failure to comply will be cause for rejection of the production since the previous test.

4.5.8.2 Bases - One hard coated base per batch per tank shall be subjected to the thickness test prescribed in the applicable specification referenced on the drawing. Failure to comply shall be cause for rejection of the production since the previous test. Procedure must be approved by the technical agency.

4.5.9 Torque of fiberglass wrapping to body - At a minimum one fiberglass wrapped body from each four hour's production from each spindle shall be subjected to this test to determine compliance with 3.4. The fiberglass wrapped body shall be mounted in a suitable holding device in such a manner that the body is prevented from turning. The fiberglass wrapping shall be notched to a suitable depth in as many places as necessary, 180 degrees apart. There shall be no compressive force when applying the applicable torque to the fiberglass wrapping. If any body fails to comply with the requirement, the production represented by the sample shall be rejected. An alternate test method and sampling plan may be submitted to the technical agency for approval

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4.5.10 Fiberglass content - The fiberglass wrapped bodies selected for the torque test shall also be subjected to this test. Three specimens (1" x 1") shall be removed from the test shell. The specimens shall be individually weighed on an analytical balance. The specimens shall then be placed in a furnace at 1200°F for a period of 30 minutes. The specimens shall then be cooled in a dessiccator and weighed. The heating and cooling procedure shall be repeated until the specimens attain a constant weight. The glass content is calculated by dividing the final weight by the initial weight and multiplying the result by 100. Failure of any wrapping to comply shall be cause for rejection of the production represented by the sample.

4.5.11 Eddy current inspection of base - The base, prior to machining the holes, shall be inspected for metal defects using the eddy current method. Bases failing to comply with requirements of 3.5 shall be rejected. Standards and test method shall be in accordance with drawing 8648916.

4.5.12 Proving ground ballistic tests - The following tests shall be conducted using an M199A1 Cannon tube which shall have 25 percent minimum remaining life based on current gun tube condemnation criteria. Photographic equipment shall be positioned near the muzzle of the weapon to record the condition of the projectile as its exits the tube. Chamber pressures and velocities shall be recorded for each round fired. All photographs and recovered hardware shall be examined for any noncompliance with the applicable requirements in 3.6. Range, deflection, and velocity shall be recorded for each round fired for informaion.

4.5.12.1 Excess pressure - Each projectile submitted for this test shall contain inert cargo, have its base assembled, a dummy expulsion charge, and a dummy fuze. Each projectile will be fired in a M199 Cannon with a M203 Propelling Charge conditioned to 145°F. An attempt will be made to recover the projecties fired for this test.

4.5.12.2 Zone 3 - Each projectile submitted for this test shall contain inert cargo, have its base assembled, contain a live expulsion charge, and a M577 fuze. Each projectile will be fired in either a M185 or M199 Cannon with a M3A1 Propelling Charge conditioned -40 + 5°F. Fuze function time shall be recorded. An attempt will be made to recover the projectiles fired for this test.

4.5.12.3 Service pressure (drop test) - The projectile body assembly and base with inert cargo and with lifting plug assembled will have a seven foot drop test performed prior to firing at service pressure. The projectile shall be dropped seven feet

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(lowest point of the projectile to point of impact) onto a steel plate that is solidly supported on concrete. The orientation of the projectile shall be at an axis of 45 degrees with the base down. The projectile shall be attached by means of a chain or cable to the lifting plug to prevent the projectile from tumbling after impact. The armor plate upon which impact occur shall have a minimum thickness of 3 inches, a Brinel Hardness of 207 minimum and shall be solidly supported in a horizontal plane over its entire bearing surface by a minimum thickness of 24 inches of concrete. The steel plate shall have a smooth flat surface. Upon completion of the seven foot drop test, the lifting plug shall be removed and the applicable fuze and expulsion charge shall be assembled into the projectile. A M203 Propellant Charge conditioned to 70°F shall be used. Otherwise, testing shall be as specified in 4.5.12.2

5. PACKAGING

5.1 Packing - This section (Section 5) is not applicable if LAP operations are to be performed at the same facility.

5.1.1 Level A - Not Applicable.

5.1.2 Leve B - Not Applicable

5.1.3 Level C - Minimum Military Pack

5.2 Packaging and packing

5.2.1 Projectile metal parts assemblies shall be palletized in accordance with Drawing 8837839.

5.2.2 Bases shall be packaged and packed in accordance with MIL-STD-1169, and will conform to contractor drawings.

5.3 Marking - Marking of the pallet shall comply with Dwg. 8837839. Marking of the cartons for the bases shall comply with MIL-STD-1169 and shall include marking to assure that the base is not shipped or stored with the cavity facing upward. Bases need not be identified with projecitle lot numbers.

6. NOTES

6.1 Intended use - The projectile metal parts covered by this specification are intended for the Projectile, 155MM: HE, M692/M731.

6.2 Ordering data - See MIL-A-48078.

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6.3 Submission of inspection equipment designs for approval - See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Data requirements - Data Item Descriptions required for submission under this specification are as follows:

Paragraph	Data requirement	Applicable DID
3.2.1, 3.2.3, 3.3.1, 3.9.1, 3.9.2, 4.5.2 6.12	Procedure Report	* DI-T-5372 (Tailored)
4.5.1.1, 4.5.1.2, 4.5.1.3, 4.5.1.4 4.5.6, 4.5.10	Quality Inspection Test Demonstration & Evaluation Report	DI-R-1724
4.5.3.1, 4.5.3.2 4.5.11	Quality Inspection Report	DI-R-1721

* Delete a, b, c, d, of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits.

The above data shall be forwarded in accordance with the requirements of the contract to Commander, ARRADCOM, ATTN: DRDAR-QAR-Q, Dover, NJ 07801.

6.5 Submission of ballistic test data and data cards - In addition to the normal distribution, one copy of all ballistic data and ammunition cards shall be forwarded to Commander, ARRADCOM, ATTN: DRDAR-QAR-Q, Dover, NJ 07801.

6.6 Cold forming - The contractor is cautioned that if cold forming is utilized in processing of the M692/M731 body, those bodies which do not completely clean up in the machining operation prior to the cold forming have a greater propensity to develop circumferential tears during the forming operation particularly if areas of non cleanup are heavily scaled.

6.7 Suggested etching procedures for removing gilding metal overlay

a. Recommended. Immerse body in commercial copper stripper which is cyanide and chromate free and prepared in accordance with manufacturer's directions. In a fresh solution, an overlay layer 0.010 to 0.015 inch thick will dissolve in 10 to 14 hours. Renew solution as necessary.

b. Solution - 500 grams Chromic oxide $Cr\ O_3$ (commercial flakes) 50 ml. sulphuric acid H_2SO_4 (technical grade) 950 ml. water H_2O.

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This solution may be used by passing current through the solution (50 amperes dc) with the immersed shell as the anode and the container as the cathode. An overlay layer 0.010 to 0.015 inch thick will require approximately two to three hours to dissolve in a fresh solution. The solution can also be used without an external current source by immersing the shell for approximately 8 to 12 hours. An active solution would clean all pits.

6.8 The specified alloys, 1340 and 4140, which are compatable with the welded overlay rotating band, have been successfully used as body material. Experience to date has indicated that 1340 steel requires closer control over the welding process and thermal treatment to prevent cracks after banding. However, this does not preclude the need for similar close control over the welding process and thermal treatment with 4140 steel to be compatible with the production process.

6.9 Recommend welding operation of overlay welded band be guided by Frankford Arsenal "Process Recommendations for Banding 155MM, M483 Projectile" dated 19 April 1973. Requests for copies should be submitted to: Commander, ARRADCOM, ATTN: DRDAR-LCU-M, Dover, NJ 07801.

6.10 Steel blooms, billets, slabs and bars

6.10.1 It is the responsibility of the manufacturer of the projectile metal parts to select a steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machiner to produce the projectile to the dimensions, mechanical properites and metal integrity specified as an end item requirement.

6.10.2 The supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade (selected by the projectile manufacturer) and complying with the specific ASTM designation and tests selected by the projectile manufacturer.

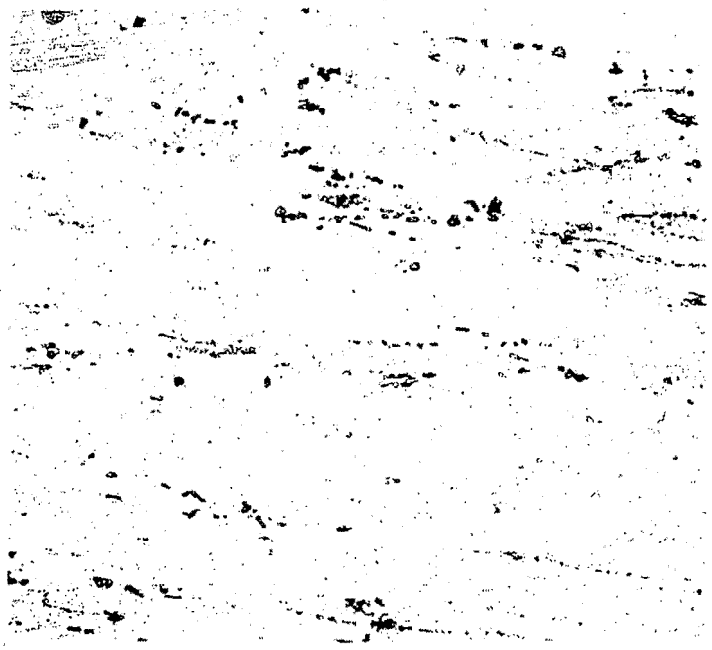
6.10.3 In the application of carbon steel, blooms, billets, slabs and bars, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions, in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment in terms of equipment sequence of operations, tooling designs, handling devices, and processing techniques having the capabilities of compensating for such variables.

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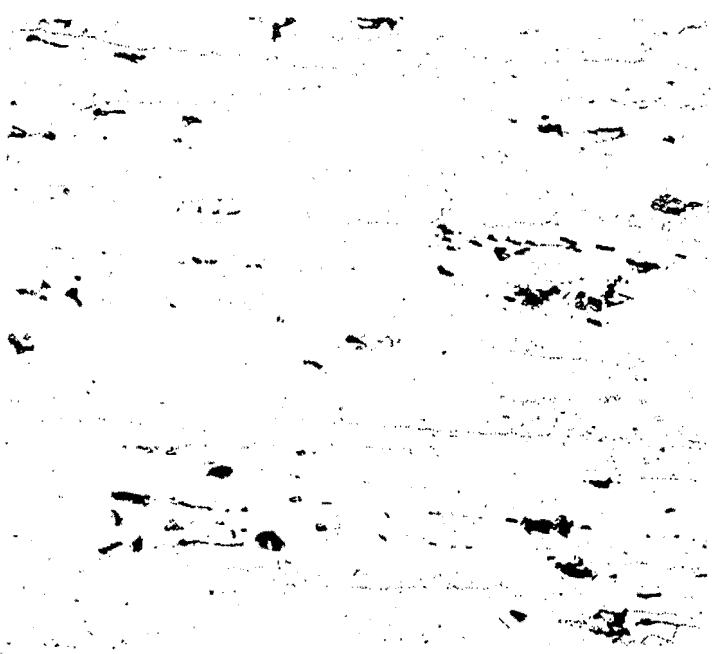
6.11 Adequate working of ogive material - In order to insure adequate working of the ogive material and the break-up of insolubles from the cast ingot to the final raw forging configuration, it is recommended that a minimum extrusion ratio of six (6) be used representing an 83% reduction of area from the cross-sectional area of the extrusion.

6.12 Heat treatment procedure - Contractor shall submit, for information purposes only, the heat treatment procedure per DI-T-5372.

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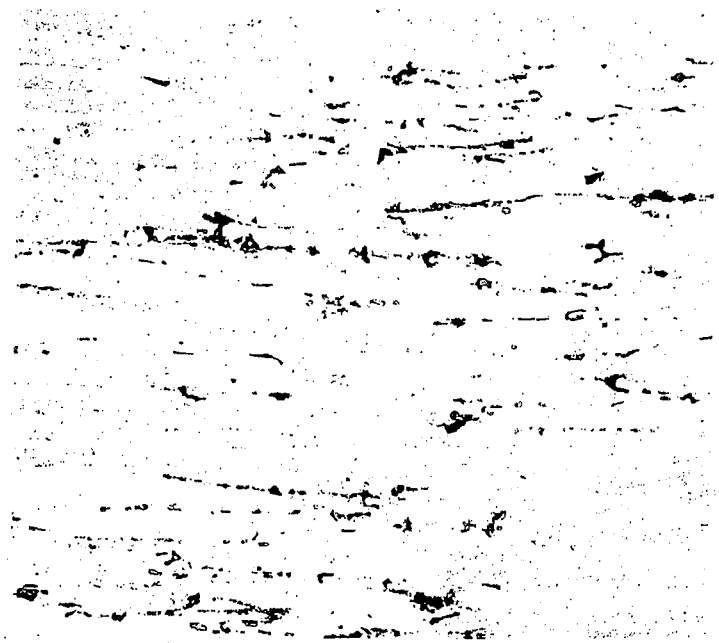
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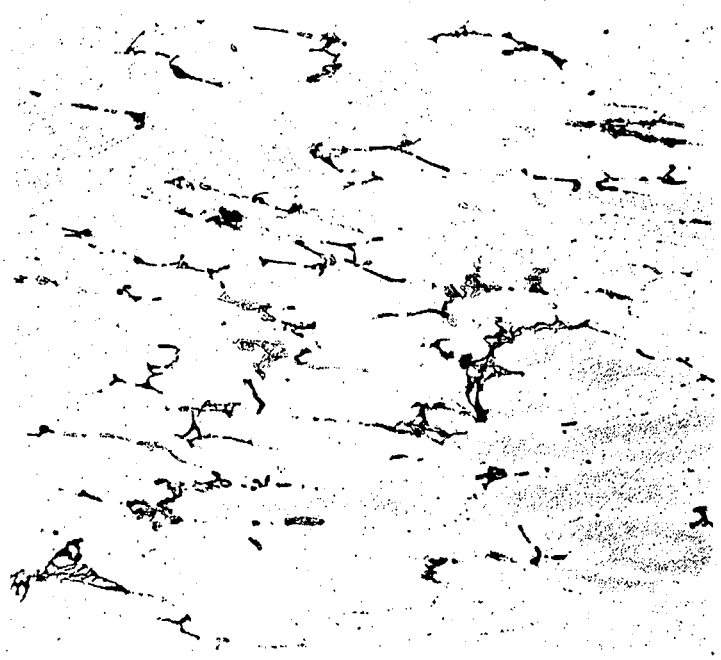
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FIGURE 1 Acceptable Microstructures (250x)

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FIGURE 2 Unacceptable Microstructures (250x)

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